

# **Ocean Color – Simultaneous Marine and Aerosol Retrieval Tool (OC-SMART)**

## **User Guide (Python)**

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## 1. Installation

### 1.1 Download and Extract OC-SMART

The [OC-SMART releases page](#) lists the versions of OC-SMART available for installation. Select the version you would like to download and then copy the link to the tar.gz file to your clipboard. Navigate to the location where you would like to download OC-SMART on your machine. Then you can download and extract the archive using the command:

```
curl -L  
https://github.com/Light-and-Life-Lab/OC-SMART/archive/refs/tags/v2.6.5.tar.gz | tar -xzf -
```

### 1.2 (Optional) Install the Conda Package Manager

Some steps in the install process are simpler if you use the Conda package manager to handle the dependencies. However, Conda is not strictly required so instructions are included for both Conda and the default Python package manager, pip.

If you are able to use Conda, the recommended distribution is Miniconda. Instructions for installing Miniconda can be found [here](#).

### 1.3 Set Up and Activate Python Virtual Environment

It is recommended that you create a fresh virtual environment in which to install OC-SMART to avoid potential version conflicts with existing Python package installations on your machine.

#### *Conda*

To set up a Conda virtual environment, run the command:

```
conda create -n <envname>
```

Then activate it using:

```
conda activate <envname>
```

Even if you are using Conda, there are certain steps in the process that require installation of Python wheels via URL. This cannot be done directly using the conda install command, so it is required that Conda users install pip within their Conda virtual environment. Once you have activated your virtual environment using conda activate <envname>, run the command:

```
conda install pip
```

This will install an instance of the pip package manager that lives inside the Conda

virtual environment.

**Note:** if you already have a different installation of Python (common for most Linux distributions, which ship with Python) then simply using `pip install <package_name>` will typically default to the version of pip that was originally installed with the OS. But for our purposes we need to use the newly installed version of pip that lives inside the Conda virtual environment. **To avoid ambiguity, all pip install <package\_name> commands used throughout this installation process should instead be `python -m pip install <package_name>`.** The `python -m` part of the command will ensure that the version of pip used for installing packages is the one installed to the currently active environment. So as long as this command is run while the Conda virtual environment is active, then the correct version of pip will be used.

### ***Pip***

To set up a regular Python virtual environment, navigate to the root of the project directory (i.e. OC-SMART) and run the command:

```
python -m venv <envname>
```

Then activate it using:

```
source <envname>/bin/activate
```

## **1.4 Install Regular Python Dependencies**

OC-SMART has a number of dependencies, most of which can be pip installed via the `requirements.txt` file. There are two additional dependencies that are not available on the pip package index and so must be installed manually via Python wheels.

### ***Conda and Pip***

For both Conda and pip this step is the same. A `requirements.txt` file is provided in the root directory of the OC-SMART repository. From this directory, install these dependencies using the following command:

```
python -m pip install -r requirements.txt
```

## **1.5 Install GDAL**

### ***Conda***

GDAL is registered to the conda-forge channel, so installing is straightforward. Simply run the command:

```
conda install conda-forge::gdal
```

### ***Pip***

a. The GDAL library can be installed via your OS's package manager. Please follow the instructions for your OS/Distribution:

*Debian/Ubuntu (including WSL):*

```
sudo apt update  
sudo apt install -y gdal-bin libgdal-dev
```

*RHEL/Fedora/CentOS:*

```
sudo dnf install -y gdal gdal-devel
```

*macOS:*

```
brew install gdal
```

b. Install the matching Python GDAL bindings:

```
pip install "GDAL==$(gdal-config --version)"
```

c. Troubleshooting

If installed successfully, the command

```
gdal-config --version
```

will print the GDAL version number. Some versions of GDAL are not available on the Python Package Index (PyPI) and so the above steps will fail. If this is the case for your version, check the available versions at

<https://pypi.org/project/GDAL/#history> and install the closest match to the version produced by `gdal-config --version`. For example:

```
pip install "GDAL==3.8.4"
```

## **1.6 Install l8angles**

The l8angles library is a Python wrapper around a C-based USGS Landsat 8 tool for computing per-pixel solar and sensor azimuth and zenith angles from Angle Coefficient Files. A set of Python wheels has been generated for the platforms supported by OC-SMART, which may be found here: [l8-angles](#). Follow the installation instructions on the linked page in order to install the appropriate Python wheel for your platform.

## 1.7 Install Atmospheric Gas Correction Library

A C++-based library for computing gas transmittance values is also used by OC-SMART. This library provides a Python interface and may be imported and used as a Python module for seamless integration with Python workflows. A set of Python wheels has been generated for the platforms supported by OC-SMART, which may be found here: [Atmospheric Gas Correction Library](#). Follow the installation instructions on the linked page in order to install the appropriate Python wheel for your platform.

## 1.8 Windows

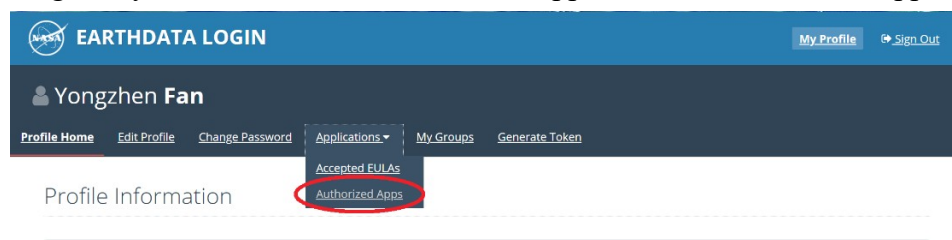
Native Windows wheels are not built or supported. Windows users should install and run OC-SMART from within [WSL \(Windows Subsystem for Linux\)](#), then follow the Linux and macOS installation instructions above from inside your WSL environment.

## 2. Ancillary data downloading

OC-SMART needs to download ancillary data from NASA OB.DAAC, the user must have an account on [earthdata.nasa.gov](https://earthdata.nasa.gov) and approve the application to download ancillary data.

**2.1** If you don't have an account on Earthdata please go to <https://urs.earthdata.nasa.gov/>, click "Register" and follow the instruction to create your account.

**2.2** Login to your Earthdata account, click "Applications->Authorized Apps",



on the next page, click "Approve more Applications" at the bottom,

**EARTHDATA LOGIN** My Profile Sign Out

**Yongzhen Fan**

Profile Home Edit Profile Change Password Applications My Groups Generate Token

### Approved Applications

Applications that use your Earthdata Login profile for authentication.

|                              |  |
|------------------------------|--|
| OB.DAAC Modis                |  |
| Earthdata Feedback Module    |  |
| Earthdata Code Collaborative |  |
| Earthdata Website            |  |
| Metadata Management Tool     |  |
| CMR SSO APP for EDL in PROD  |  |
| LAADS Web                    |  |

**APPROVE MORE APPLICATIONS**

then search for “OB.DAAC” and check “Show applications that can be auto-authorized”, then click “Authorize” on “OB.DAAC Data Access”.

**EARTHDATA LOGIN** My Profile Sign Out

**Yongzhen Fan**

Profile Home Edit Profile Change Password Applications My Groups Generate Token

### Approve Applications

OB.DAAC **SEARCH**

#### Application Results

These applications have a EULA, and must be authorized before you can use them

|                     |                  |
|---------------------|------------------|
| OB.DAAC Data Access | <b>AUTHORIZE</b> |
| OB.DAAC MERIS       | <b>AUTHORIZE</b> |
| OB.DAAC Sentinel    | <b>AUTHORIZE</b> |

☒ Show applications that can be auto-authorized

## 2.3 set up automatic authentication for data downloading

### 2.3.1 For Linux users:

- 1) In OC-SMART package, locate the files “.netrc.template” and “.urs\_cookies.template” in the ‘DataDownloadAuthentication/Linux’ directory.
- 2) Rename these two files to remove the “.template” suffix (i.e. they should be named “.netrc” and “.urs\_cookies”.
- 3) Replace 'USERNAME' and 'PASSWORD' in the “.netrc” and “.urs\_cookies”

file with your own Earthdata login credentials.

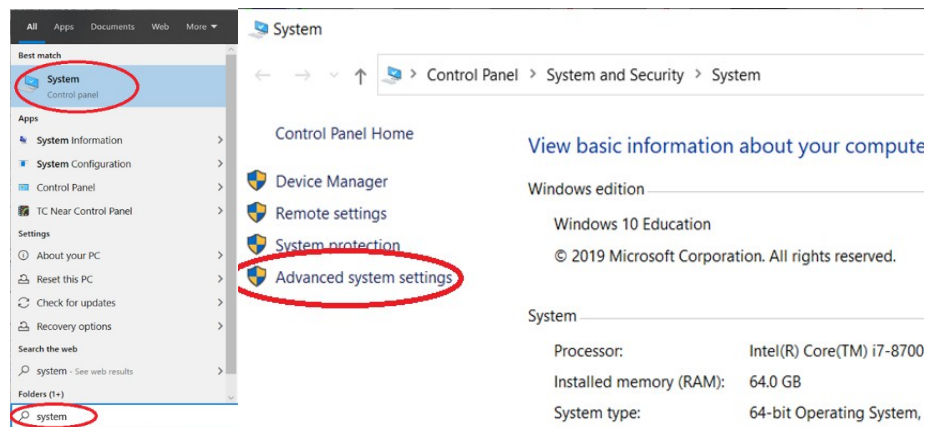
- 4) Copy the “.netrc” and “.urs\_cookies” file to your home directory, i.e. /home/<user account>
- 5) Open terminal, go to your home directory and run command:  
`chmod 0600 ~/.netrc`

### 2.3.2 For Windows users:

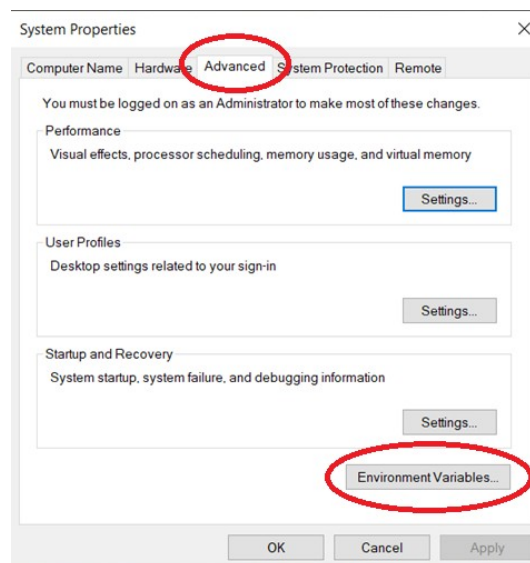
- 1) Setup “HOME” environment variable

**WARNING: change the environment variable may jeopardize your system. If you feel you are not qualified to do so, please consult with your system administrator.**

- a) Search “system” and open System(control panel), then click “Advanced system settings”



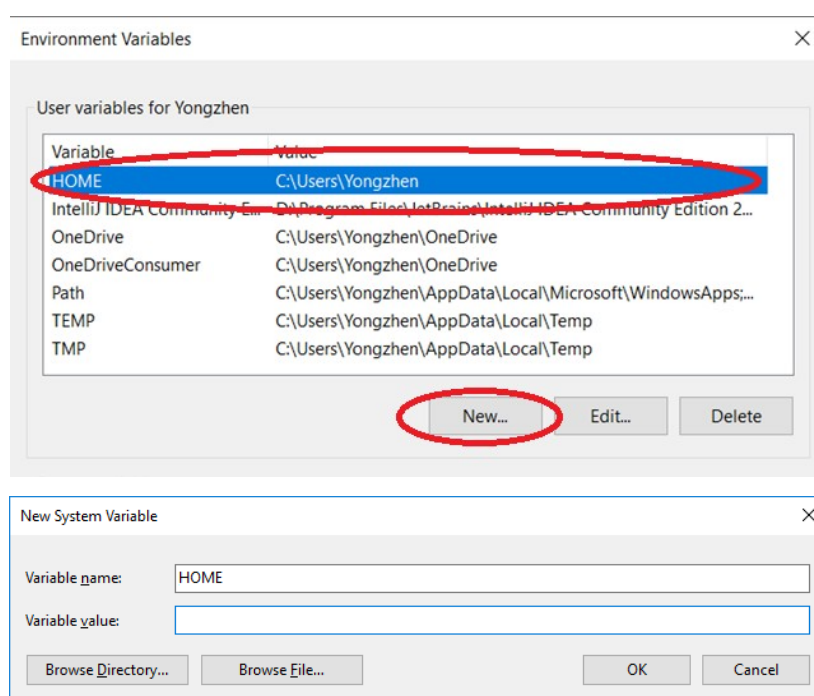
- b) Click on “Advanced” tab and then open “Environment Variables...”



c) Create “HOME” variable

**WARNING: if “HOME” variable already exists in the variable list, please DO NOT CHANGE it, as it may jeopardize other apps in your system, simply write down the path where the “HOME” variable specified and use it in step 4).**

If the “HOME” variable does not exist in the variable list, you may create the variable by clicking “New...” and input “HOME” in “Variable name” and input the path you would like to be your “home director” in “Variable value”, for example “C:\Users\<user name>”.



- 2) In OC-SMART package, locate the file “\_netrc” in the “DataDownloadAuthentication/Windows” directory.
- 3) Replace 'USERNAME' and 'PASSWORD' in the “\_netrc” file with your own Earthdata login credentials.
- 4) Copy the “\_netrc” file to the location where the “HOME” variable specified.

### 3. How to run OC-SMART

#### 3.1 System memory requirement

We recommend a minimum of 16GB of physical system memory to run OC-SMART, 64GB is recommended for best performance on larger images, such as GOCI, Landsat-8, Sentinel-2 and Sentinel-3. If physical memory is limited on your



system, please increase the size of the virtual memory, i.e., the pagefile in Windows system or the swap file in Linux system.

### 3.2 Run OC-SMART

After downloading and unzip OC-SMART, you may run OC-SMART in command line or using a Python IDE, such as Spyder.

To run OC-SMART in command line, open terminal in Linux system or Anaconda Prompt in the Windows system, navigate to the OC-SMART directory then run command:

```
python OCSMART.py
```

To run OC-SMART in a Python IDE, such as Spyder, open OCSMART.py in the IDE then click run.

### 3.3 Supported sensor

SeaWiFS, MODIS (Aqua and Terra), VIIRS (SNPP), GOCI (COMS), SGLI (GCOM-C), OLCI (Sentinel-3A), MSI (Sentinel-2A and Sentinel-2B), EPCI (DSCOVR), MERISI-II (FY-3D), HICO (ISS), OLI (Landsat-8, **NOT supported in Linux version**).

### 3.4 Input data files

OC-SMART requires the level-1B(L1B) satellite reflectance data and the associated geolocation file as input. L1B data file for SeaWiFS can be acquired by processing the L1A data using NASA SeaDAS software.

The L1B satellite reflectance data file from all supported sensors can be located in a common directory. Some sensors, like MODIS and FY-3D MERISI-II have separated geolocation files, which **MUST** be located in a different directory. An example of the L1B and GEO directory may look like:

---

DATA2 (E:) > Data2 > OCSMART\_webversion > OCSMART\_TestingData > L1B >

---

-  LC08\_L1TP\_013032\_20200309\_20200314\_01\_T1
-  S2A\_MSIL1C\_20180512T015701\_N0206\_R060\_T52SFB\_20180512T035244.SAFE
-  S2B\_MSIL1C\_20180427T015649\_N0206\_R060\_T52SFB\_20180427T035721.SAFE
-  S3A\_OL\_1\_EFR\_\_20170422T015825\_20170422T020125\_20170423T062739\_0180\_017\_003\_2339\_LN1\_O\_NT\_002.SEN3
-  COMS\_GOCI\_L1B\_GA\_20140503031644.he5
-  epic\_1b\_20180523055430\_02.h5
-  FY3D\_MERSI\_GBAL\_L1\_20190313\_0440\_1000M\_MS.HDF
-  GC1SG1\_201812170231N07110\_1BSG\_VNRDL\_1001.h5
-  H2014072171335.L1B\_ISS.nc
-  MYD021KM.A2015112.0500.061.2018049095205.hdf
-  NPP\_VMAES\_L1.A2016042.1912.001.2017142190920.hdf
-  S2002011040808.L1B

DATA2 (E:) > Data2 > OCSMART\_webversion > OCSMART\_TestingData > GEO

 FY3D\_MERSI\_GBAL\_L1\_20190313\_0440\_GEO1K\_MS.HDF  
 MYD03.A2015112.0500.061.2018048161835.hdf

### 3.5 Input parameter setup

The input parameters of OC-SMART can be set in the file *OCSMART\_Input.txt*.

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| l1b_path                | Directory where L1b data is located                             |
| geo_path                | Directory where geolocation data is located                     |
| l2_path                 | Directory where L2 output data will be located                  |
| l2_prod                 | List of products to be written to the L2 file                   |
| solz_limit              | Maximum solar zenith angle (must be $\leq 70$ )                 |
| senz_limit              | Maximum sensor viewing angle (must be $\leq 70$ )               |
| gzip_compression_opt    | Option for L2 file writing compression                          |
| floating_point_datatype | Determines size of floating point numbers used for calculations |

#### 3.5.1 Input Parameter Accepted Values

- **l1b\_path, geo\_path, l2\_path:** Formatted like a Linux path (i.e. with forward slashes). Using “.” will resolve to the directory containing OCSMART.py.
- **solz\_limit, senz\_limit:** A floating point number, must be less than 70.0. If these options are not listed in OCSMART\_Input.txt, OC-SMART uses a value of 70.0 by default.
- **floating\_point\_datatype:** Options are “float32” and “float64”. Determines the size of floating point values used in the large arrays used by OC-SMART. Using “float32” reduces memory usage, runtime, and output file size.

Output using both modes has been tested, and using float32 does not incur significant error accumulation compared to using float64. For example, using float32 for a PACE input file produces L2 product values with relative error on the order of  $10^{-5}$  or smaller.

#### 3.5.2 The following example will process the entire satellite image:

```

l1b_path = ./L1B/
geo_path = ./GEO/
l2_path = ./L2/
solz_limit = 70.0
senz_limit = 70.0
floating_point_datatype = float32

```

### 3.5.3 Sub-image processing

There are 3 options to define a sub-image in OC-SMART. You need to add following parameters for each option.

a) Define sub-image using a range of latitude and longitude. OC-MART will extract the sub-image defined by the north-west and south-east corner. To use this option, add following parameters in *OCSMART\_Input.txt*

```

north = <max latitude>
south = <min latitude>
east = <max longitude>
west = < min longitude >

```

Note: use negative values for latitude in southern hemisphere and longitude in western hemisphere.

b) Define sub-image using center latitude/longitude and the size of a box. OC-SMART will find the pixel that is closest to the input center latitude/longitude and extract a sub-image in the same size as user defined. To use this option, add following parameters in *OCSMART\_Input.txt*

```

latitude_center= <latitude>
longitude_center = < longitude >
box_width = <number of pixels>
box_height = < number of scanlines >

```

NOTE: i) use negative values for latitude in southern hemisphere and longitude in western hemisphere. ii) *number of pixels* and *number of scanlines* should be an odd number, if even numbers are given, the size of the output is (*number of pixels*+1) by (*number of scanlines*+1). iii) if the input center latitude/longitude is very close to the edge of the satellite image, OC-SMART only extract the part of the sub-image that is overlapped with the satellite image.

c) Define sub-image with range of the scanline and pixel number. To use this option, add following parameters in *OCSMART\_Input.txt*

*start\_line* = <start scanline number>  
*end\_line* = < end scanline number >  
*start\_pixel* = < start pixel number >  
*end\_pixel* = < end pixel number >

Note: i) input scanline and pixel number should be ZERO based. ii) This option can be used to process whole image in blocks if system memory is limited.

### 3.6 Output file

OC-SMART output level-2 (L2) product in HDF5 format. The output is compatible with NASA SeaDAS and ESA SNAP for visualization, and can be read by other software that support HDF5 format, such as Matlab, HDFView, Python, etc.

#### 3.6.1 The L2 products include:

- Spectral aerosol optical depth (AOD)
- Spectral normalized remote sensing reflectance (Rrs)
- Spectral total absorption by particulates and CDOM in water (at)
- Spectral total scattering by particulates in water (bt)
- Spectral absorption by Phytoplankton (aph)
- Spectral absorption by detritus and gelbstoff (adg)
- Spectral backscattering by particulates (bbp)
- Chlorophyll\_a concentration by NASA OCi algorithm (chlor\_a(oci))
- Chlorophyll\_a concentration by YOC algorithm (chlor\_a(YOC))
- Total suspended matter by YOC algorithm (tsm(YOC))
- Gas transmittance values along solar zenith and sensor zenith paths:
  - *tg\_sol\_<gas>*, *tg\_sen\_<gas>* for eight molecular species (i.e. gas = O<sub>3</sub>, CO<sub>2</sub>, CO, CH<sub>4</sub>, NO<sub>2</sub>, N<sub>2</sub>O, O<sub>2</sub>, H<sub>2</sub>O)
  - *tg\_sol*, *tg\_sen* (product of all individual gas transmittances)
- Solar zenith angle, sensor zenith angle and relative azimuth angle
- Latitude and Longitude
- L2\_flags

#### 3.6.2 L2\_flags

- 0: Valid pixel
- 1: Satellite L1 reflectance unavailable (i.e., saturation or missing values)
- 4: Solar or sensor viewing angle out of range
- 16: Land
- 64: Cloud
- 256: Rayleigh corrected reflectance (Lrc) out of scope
- 1024: negative Rayleigh corrected reflectance (Lrc)

### **3.7 clean temporary files**

OC-SMART will save ancillary files to your local drive, if storage space is limited, manually delete the ancillary files located in `./anc/` and `./landmask_gsw/`.

## **4. Referencing**

When acknowledging the use of OC-SMART for scientific papers, reports etc. please cite the following reference:

*Fan, Y., Li, W., Chen, N., Ahn, J., Park, Y., Kratzer, S., Schroeder, T., Ishizaka, J., Chang, R., and Stamnes, K., (2020) "OC-SMART: A machine learning based data analysis platform for satellite ocean color sensors", Remote Sens. of Environ., Vol. 253, p11236, DOI: 10.1016/j.rse.2020.112236.*

## **5. Term of Use**

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